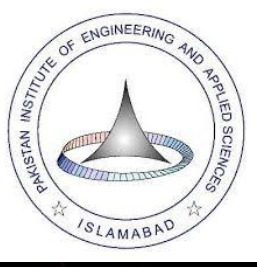


Quantum Algorithms for Optimization of Constraint Problems



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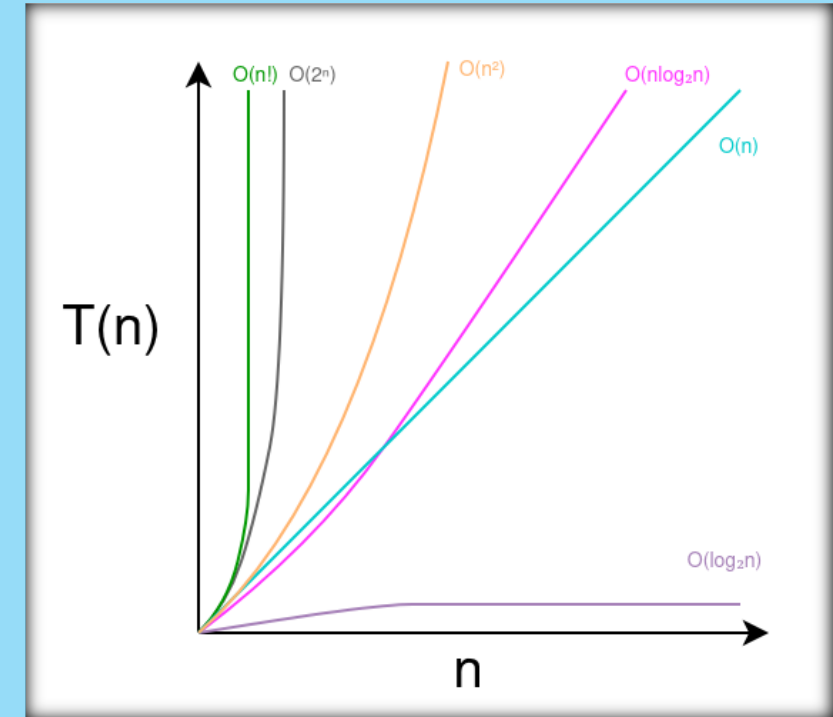


Contents

- Motivation
- QAOA overview
- Constraint Optimization
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- Conclusion and Future Direction

Motivation

- Combinatorial Problems are NP-Hard
- Example: TSP , MaxCut etc.
- Classical Complexity is **EXPONENTIAL**
- Supercomputers even fails for large instances
- Here, Quantum Computers offer new paradigms



Time Complexity Comparison of Problems



QAOA Overview

(Quantum Approximate Optimization Algorithm)

- **QAOA:** A hybrid (Quantum-Classical) Algorithm
- **Key Idea:** Prepare the parametrized quantum state by alternating
 1. *Cost unitary:* encodes the objective function ($e^{-i\gamma H_c}$)
 2. *Mixer unitary:* explore the complete solution space ($e^{-i\beta H_M}$)
- **Measurement:** Measure Expectation value and update the parameters with optimizer till convergence

Constraint Optimization



Soft Constraints

- These constraints should be respected.
- Enforced by penalties in the cost Hamiltonian.



Hard Constraints

- These constraints must be respected.
- Infeasible states are never explored by the mixer Hamiltonian.



Constraint Optimization

- Standard QAOA inefficient for handling constraints
- Two approaches widely used

1. Penalty Based QAOA

$$\mathbf{H}_C = \mathbf{H}_c + \lambda \mathbf{H}_{\text{penalty}}, \quad \mathbf{H}_M = \sum_{i=1}^n X_i$$

2. Constraint Preserving QAOA

$$\mathbf{H}_C = \mathbf{H}_c, \quad \mathbf{H}_M = \mathbf{H}_{XY} = \frac{1}{2} \sum_{i,j}^n (X_i X_j + Y_i Y_j)$$



Weighted MaxCut

- Well known problem in ***Graph Theory*** and ***CO*** since 1970
- First time formulated by **Richard Karp** in 1972
- “*Partitioning the vertices of graph into two disjoint set to maximize the weights of the edges between these sets.*”
- **Edward Farhi** in November 2014 first time apply QAOA on MaxCut

An aerial photograph of a complex highway interchange with multiple overpasses and ramps. The roads are filled with a dense flow of vehicles, including cars and trucks, creating a colorful mosaic of traffic. The surrounding area includes some commercial buildings and parking lots. The word "Traffic" is superimposed in the center in a white serif font.

Traffic



Traffic Network Partitioning

- Map Traffic Network as **Weighted MaxCut Problem** $G(V,E)$
- Nodes represent the **Intersections**
- Edges represent the **Roads**
- Weights represent the **Traffic Intensity**
- **Goal**
“ Partition the network in two disjoint regions while optimizing the overall traffic distribution ”
- **Constraint**
“ Here in my specific problem is balance constraint. ”

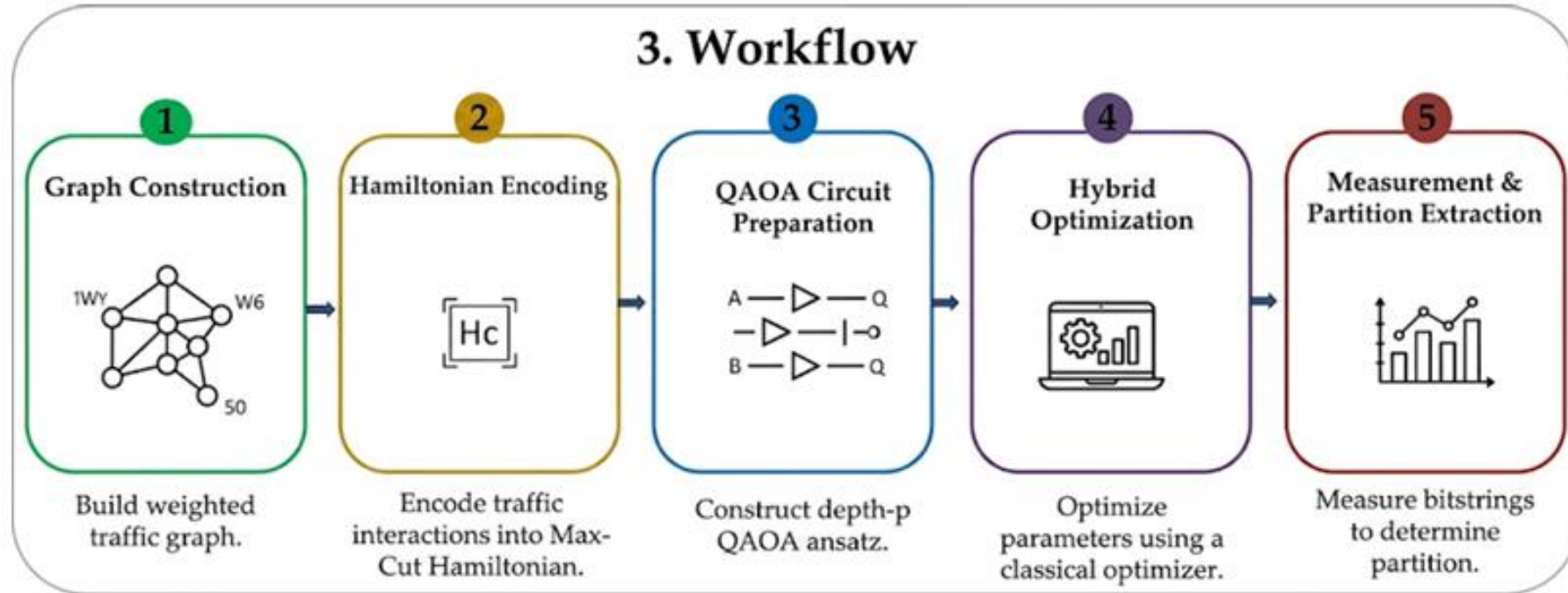
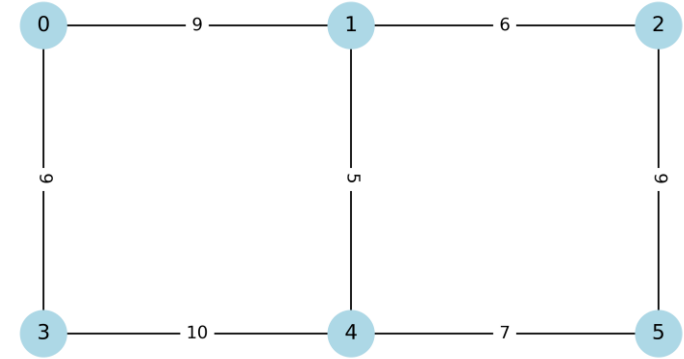


FIGURE 6.1: *Schematic overview of the proposed five-stage optimization workflow: (1) mapping the structural data into a weighted traffic graph; (2) encoding network routing constraints and interactions into a problem Hamiltonian; (3) preparing the parameter-dependent quantum circuit ansatz; (4) executing the hybrid classical-quantum optimization loop; and (5) measuring final bitstrings to decode the optimal network partition.*

Traffic Network Partitioning

- Toy model with 6 intersection and 7 roads $G(6,7)$
- **Penalty Based QAOA**

$$H_C = -\frac{n\lambda}{2} + \sum_{i,j} \left(\frac{w_{ij}}{2} + \frac{\lambda}{2} \right) Z_i Z_j$$



qaoa_maxcut_2x3_grid.html

- **Constraint Preserving QAOA**

$$H_M = \sum_{i<j} (X_i X_j + Y_i Y_j).$$

$$|\psi(\gamma, \beta)\rangle = \prod_{l=1}^p e^{-i\beta_l H_M} e^{-i\gamma_l H_C} |+\rangle^{\otimes n}.$$

Results

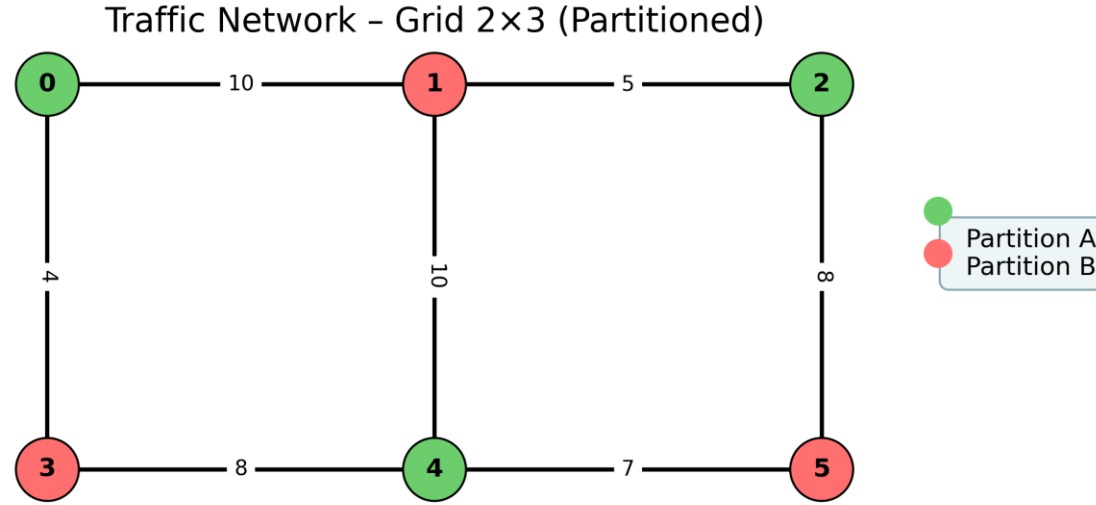
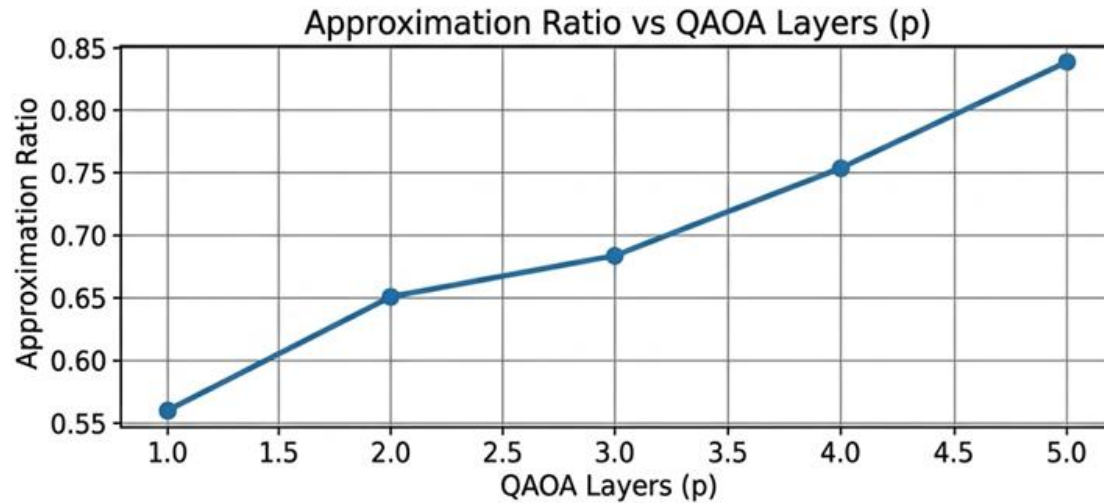


TABLE 6.1: *Performance Comparison of Different Quantum Approximate Optimization Algorithm (QAOA) Variants for Traffic Network Partitioning.*

Depth (p)	Algorithmic Approach	Cut Value	Approximation Ratio
$p = 1$	Penalty RQAOA	55.00	1.0000
$p = 3$	Constraint-Preserving Mixer	52.13	0.9478
$p = 5$	Penalty QAOA	45.88	0.8341

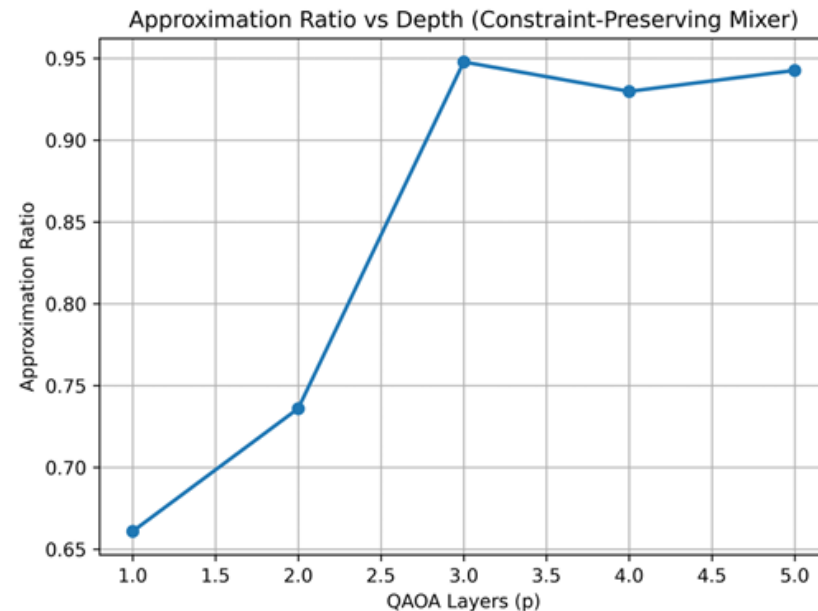
Results

Penalty Based QAOA



The bottom panel displays the approximation ratio achieved as a function of the QAOA layers, p . The plot clearly shows that increasing the number of layers leads to a consistent improvement in the final approximation ratio, demonstrating the efficacy of greater depths in the QAOA algorithm.

Constraint Preserving QAOA



Approximation ratio as a function of QAOA circuit depth ($p \in \{1, 2, 3, 4, 5\}$) utilizing the constraint-preserving mixer strategy. The performance scales sharply from an initial ratio of ≈ 0.66 at $p = 1$ to a peak performance near 0.95 at $p = 3$, demonstrating the direct impact of ansatz depth on overcoming local minima when optimizing within the feasible subspace.

Results



Scaling of Balanced Traffic Network Partitioning using QAOA

Number of Nodes (N)	Number of Balanced Partitions	Circuit Complexity (Approx. Total Gates)
4	6	66
8	70	132
10	252	165
20	184,756	330
30	155,117,520	495
40	137,846,528,820	660
50	126,410,606,437,752	825

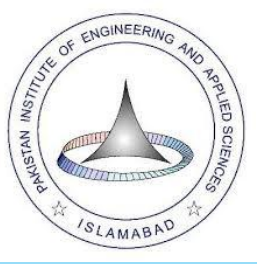
Conclusion

- Traffic Network Partitioning was formulated as a constrained weighted Max-Cut optimization problem.
- The balance constraint was incorporated using two approaches:
 - Penalty-Based QAOA*
 - Constraint-Preserving QAOA*
- QAOA was successfully applied to obtain feasible partitions while satisfying the problem constraints.
- The study demonstrates the potential of quantum optimization algorithms for solving constrained combinatorial optimization problems.
- The proposed framework establishes a bridge between graph partitioning problems and real-world traffic network applications.



Future Direction

- Extend the study to larger and more realistic urban traffic networks.
- Investigate additional practical constraints such as road capacity, traffic demand, and multi-region partitioning.
- Implement the proposed approaches on real quantum hardware and compare their performance with classical simulations.
- Study the impact of quantum noise and hardware limitations on solution quality.



Extra Slides



Classical Optimizers

- COBYLA (constrained optimization by linear approximation)
Select trust region from solution space → Linear Approximation in that trust region
- SPSA (simultaneous perturbation stochastic approximation)
Approximate the gradient by varying all parameters simultaneously
- SLSQP (sequential least square programming)
For optimization of complex parameters

Penalty Based QAOA

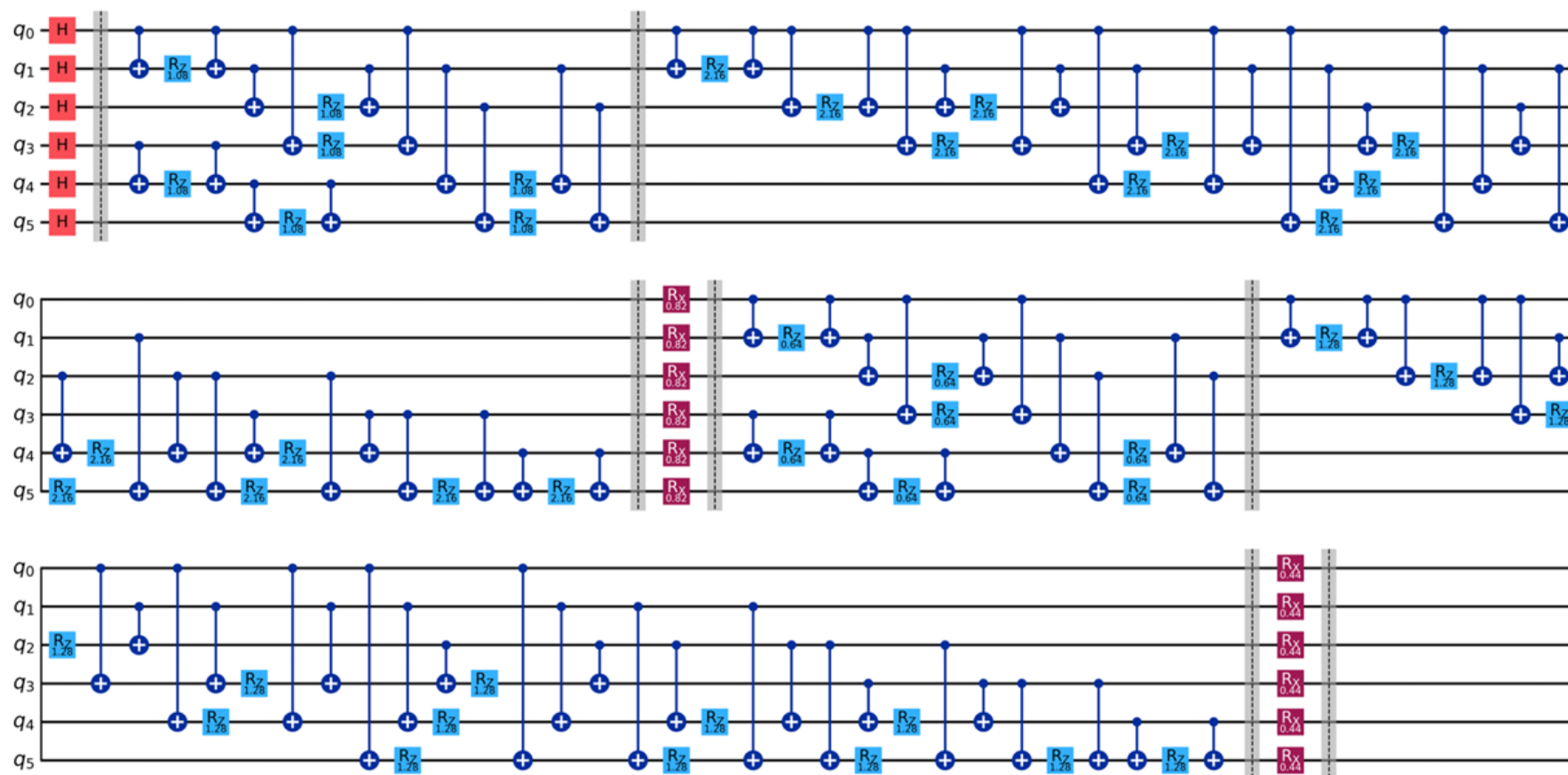


FIGURE 6.3: Gate sequence diagram for the 6-qubit penalty-based QAOA formulation, mapped across lines q_0 through q_5 .

Constraint Preserving OAOA

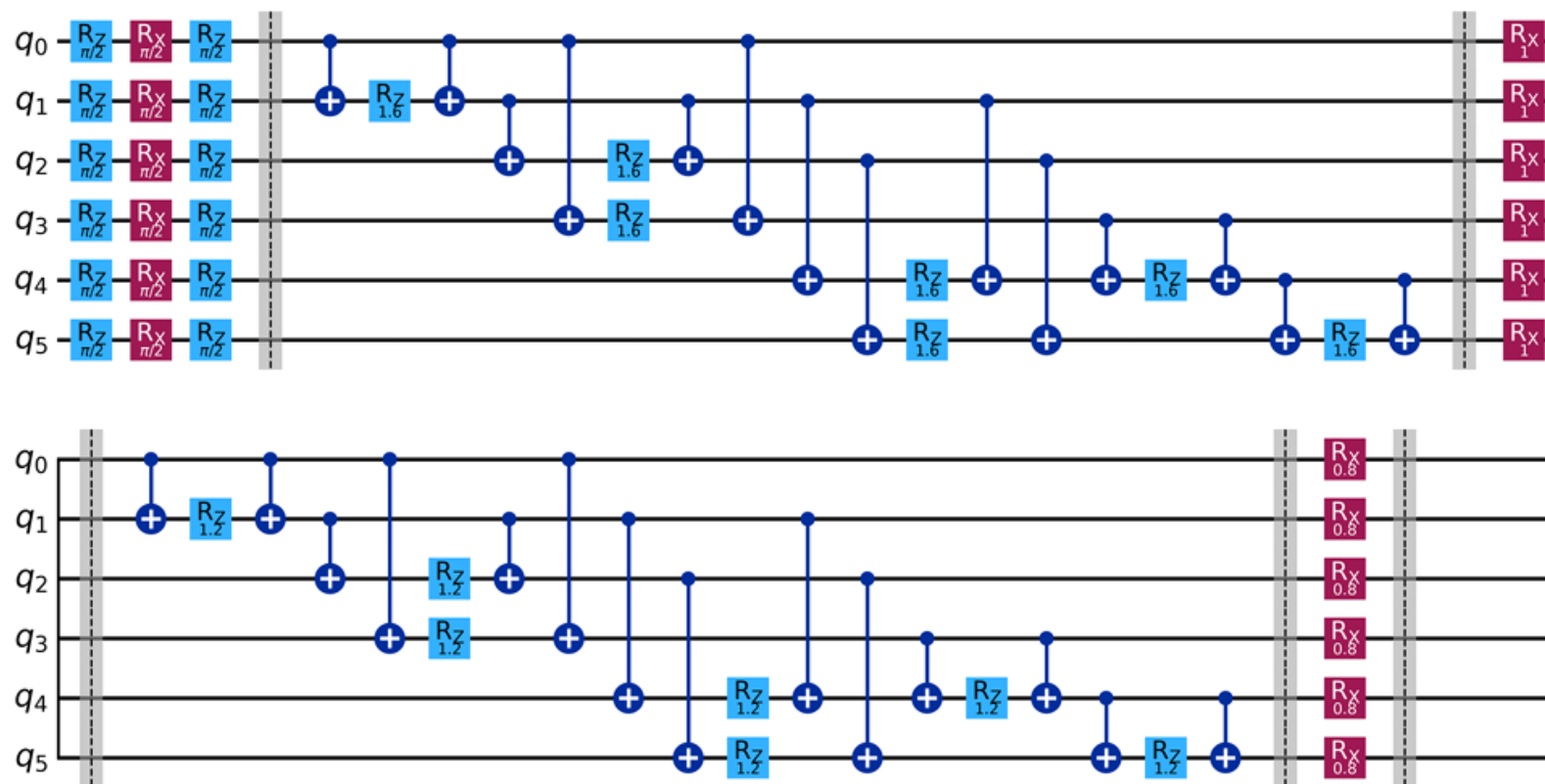
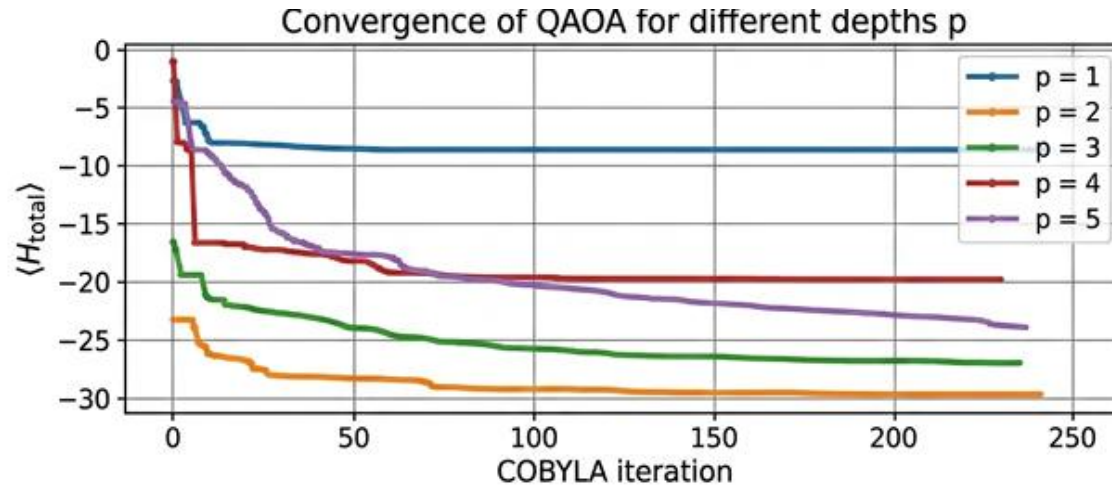
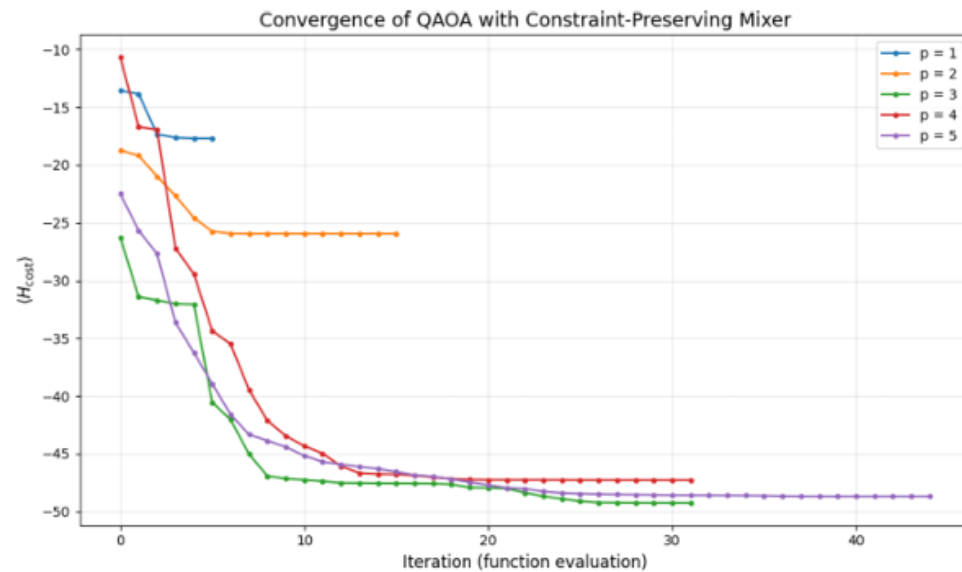


FIGURE 6.5: Gate-level circuit schematic for the 6-qubit mixer-based QAOA formulation showing the complete initialization and variational block sequence for a 2×3 grid network.

Some Other Results



The top panel shows the convergence of the expected total Hamiltonian, $\langle H_{total} \rangle$, as a function of COBYLA optimization iterations for various circuit depths, p . For each depth $p \in \{1, 2, 3, 4, 5\}$, $\langle H_{total} \rangle$ decreases with subsequent iterations, with greater depths generally leading to lower expectation values.



Convergence profiles of the expectation value of the cost Hamiltonian $\langle H_{cost} \rangle$ as a function of classical optimizer iterations (function evaluations). The optimization tracks the performance of the constraint-preserving mixer variant across varying circuit depths from $p = 1$ to $p = 5$, demonstrating deeper ansatzes reaching lower energy minima at the expense of longer optimization paths.

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**THANK
YOU**